PHASE 3 PROJECT

Auto Insurance

Fraud Detection

Using Machine Learning to protect Car_Nzuri Auto Insurance and its policyholders from fraudulent claims

THE PROBLEM WE ARE SOLVING

$40B+

Lost to insurance fraud annually in the US

~15%

Higher premiums paid by honest policyholders

1 in 4

Claims in our dataset was fraudulent

Our Goal

Build a machine learning model that automatically scores every incoming claim on its likelihood of fraud — so investigators can focus their limited time on the cases that matter most, rather than reviewing thousands of claims manually.

BUSINESS & DATA UNDERSTANDING

The Dataset

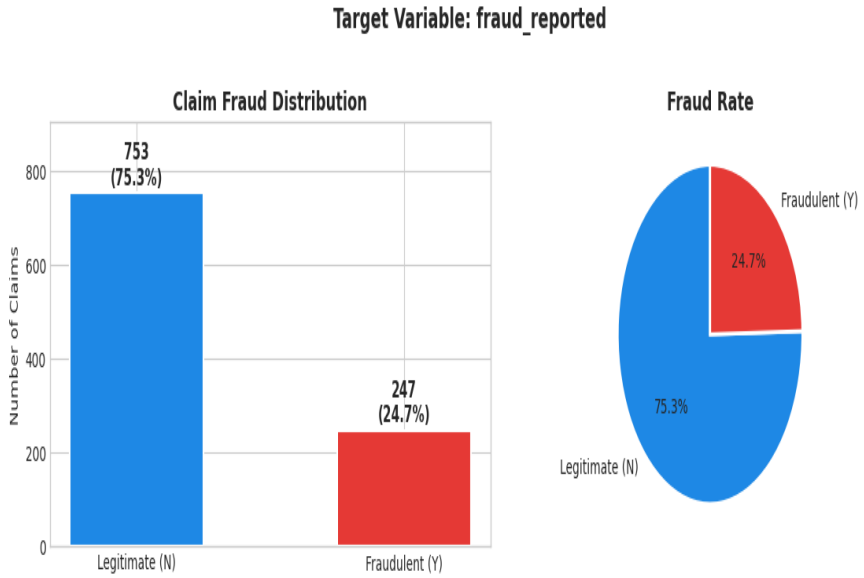
1,000 claims
real auto insurance records

39 features
per claim across 4 categories

25% fraud rate
moderate class imbalance

Car_Nzuri
Auto Insurance Company

Fraud vs. Legitimate Claims



WHY MACHINE LEARNING?

❌ Without ML

- Investigators manually review every single claim
- Easy to miss patterns hidden in thousands of records
- Fraud slips through undetected — company loses money
- Good customers face higher premiums as a result

→

✅ With ML

- Every claim automatically receives a fraud risk score
- Model spots subtle patterns humans would miss
- Investigators focus only on the highest-risk cases
- Faster payouts for honest customers

OUR MODELLING APPROACH

We tested four progressively more sophisticated approaches — each building on lessons from the last:

1

Starting Point

Baseline Model

A simple first pass to set the bar. Useful to expose the core challenge: the model ignored most fraud cases.

2

Smart Adjustment

Tuned Logistic Regression ⭐

Added fraud-aware weighting so the model stops ignoring the minority class. This was the most impactful change.

3

Rule-Based Tree

Decision Tree

Makes decisions by following a flowchart of if/then rules — very easy for non-technical staff to read and explain.

4

Ensemble Forest

Random Forest

Hundreds of trees vote together. Theoretically powerful, but overfitted on our dataset size.

EVALUATION RESULTS

🎯 **F1-Score**

Overall balance between catching fraud and avoiding false alarms

🔍 **Recall**

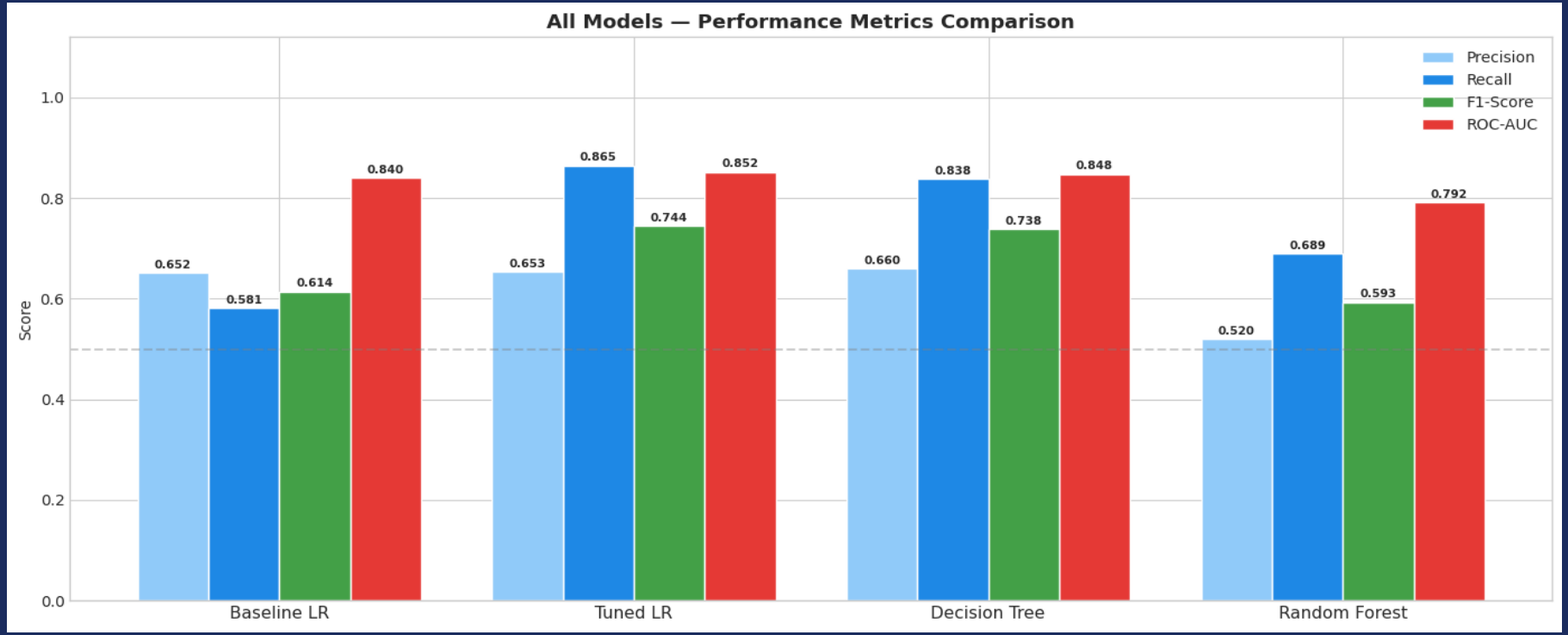
How many real fraud cases we actually caught

📊 **ROC-AUC**

How well the model ranks risky claims above safe ones

Model	Fraud Caught (Recall)	Precision	F1-Score ★	ROC-AUC	Test Acc
Baseline Logistic Regression	58%	65%	0.6143	0.8403	82%
⭐ **Tuned Logistic Regression**	**86%**	65%	**0.7442**	**0.8521**	85%
Tuned Decision Tree	84%	66%	0.7381	0.8479	85%
Tuned Random Forest	69%	52%	0.5930	0.7917	77%

HOW THE MODELS COMPARE



The Tuned Logistic Regression (blue bars) leads on every key metric — especially Recall (86%), meaning it catches the most fraud.

WHAT SIGNALS FRAUD?

The model identified four features that most strongly predict a fraudulent claim:

01 Claim Size vs. Premium Paid

When the amount claimed is very high relative to what the customer pays in premiums, that mismatch is a red flag.

02 Incident Severity

Fraudulent claims tend to involve more "major" or "total loss" incidents — extreme severity is disproportionately common in fraud.

03 Vehicle Claim Amount

Inflated vehicle repair or replacement costs are one of the most reliable fraud signals the model found.

04 Total Claim Amount

Overall, larger claims carry a statistically higher fraud risk — though not all large claims are fraudulent.

RECOMMENDATIONS

🚀 1. Deploy the Model as a Fraud-Scoring Engine

Every new claim automatically receives a risk score from 0–100. Claims above a set threshold go to the investigation queue — others are fast-tracked for approval.

🌳 2. Use the Decision Tree for Compliance & Explainability

When a claim is flagged, the Decision Tree model can produce a plain-English explanation of exactly why — essential for regulators, managers, and claimants who ask "why was my claim held?"

🎯 3. Focus Investigators on the Top 4 Signals

Claim-to-premium ratio, incident severity, vehicle claim amount, and total claim amount should be the first things every investigator checks on a flagged case.

🔄 4. Retrain the Model Every Quarter

Fraudsters adapt their tactics. Regular retraining on fresh data ensures the model stays sharp and keeps up with evolving fraud patterns.

NEXT STEPS

Short Term

1. Pilot the model on 60 days of live claims data
2. Set investigation threshold based on team capacity
3. Establish performance monitoring dashboard

Medium Term

1. Conduct formal cost-benefit threshold analysis
2. Integrate model scores into claims management system
3. Train investigators on interpreting model outputs

Long Term

1. Explore advanced models (XGBoost / LightGBM)
2. Build feedback loop from investigation outcomes
3. Quarterly model retraining and performance review

Thank You

Questions & Discussion

86%	0.74	0.85
Fraud Detected	F1-Score	ROC-AUC